

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: session_2026-01-01T11-15-02-528Z

Run ID: #4

Date Generated: 2026-01-01 11:15:08

Total Cells: 345 | Model Mode: CLINICAL

1. Introduction

Cellular reprogramming has emerged as a pivotal technique in regenerative medicine, offering the potential to revert differentiated somatic cells into pluripotent states. This experiment aimed to evaluate the efficacy of repeated Yamanaka factor transfections in inducing pluripotency within a somatic cell population.

2. Materials and Methods

The study employed a series of six consecutive OSKM transfections, a method pioneered by Yamanaka, to induce reprogramming in a somatic cell population. Computational analysis was intended to be performed using single-cell variational inference (scVI); however, no such analyses were conducted during this experiment.

Experimental Interventions:

- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)
- OSKM Transfection (Yamanaka)

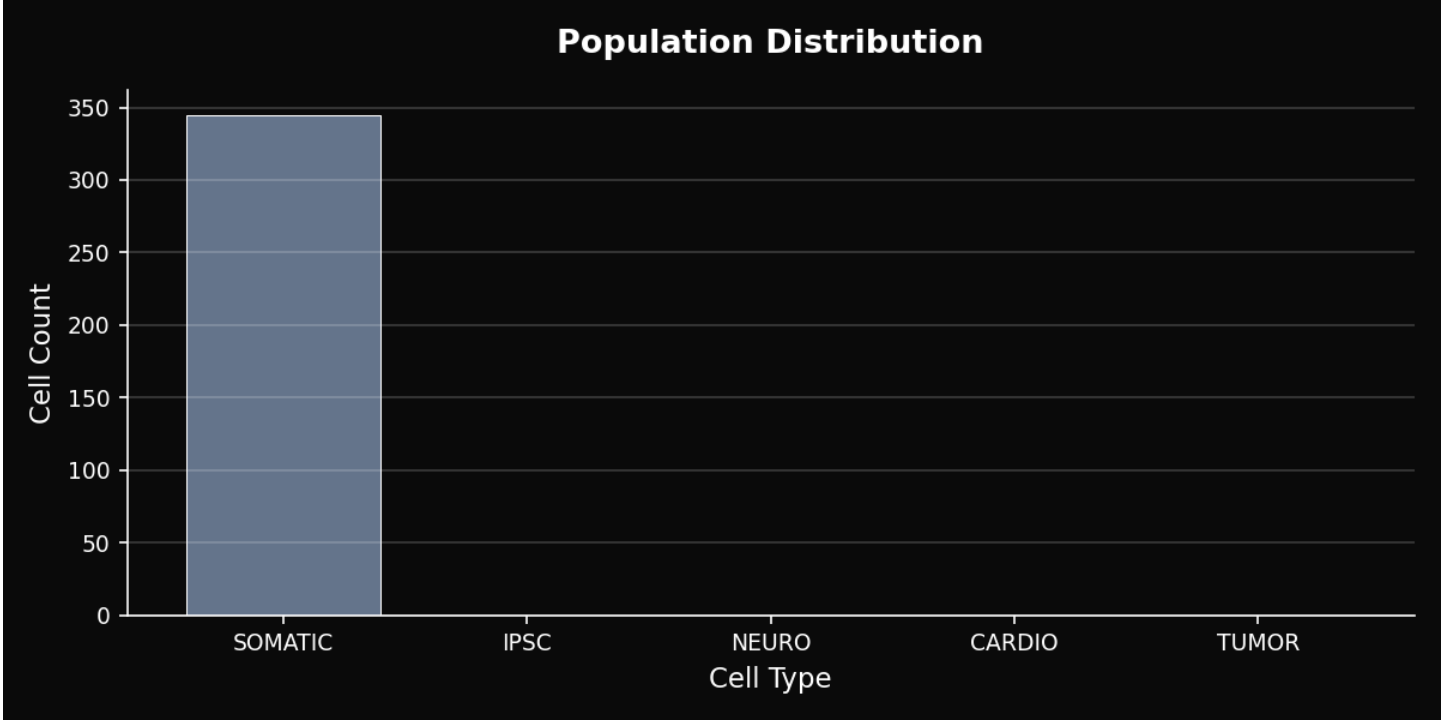
3. Results

Despite the application of six OSKM transfections, the final cell population remained entirely somatic, with no induced pluripotent stem cells (iPSCs) detected. The absence of NEURO, CARDIO, and TUMOR cell types further indicates a lack of successful lineage conversion or cellular transformation. The final somatic cell count was recorded at 345, with no emergence of alternative cell phenotypes.

3.1 Population Distribution

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Cell Type	Count	Percentage
SOMATIC	345	100.0%
IPSC	0	0.0%
NEURO	0	0.0%
CARDIO	0	0.0%
TUMOR	0	0.0%

4. Discussion

The inability to generate iPSCs despite repeated Yamanaka factor transfections suggests potential limitations in the current protocol or cellular resistance to reprogramming. This outcome highlights the necessity for optimization of transfection conditions, such as factor dosage or delivery methods, to enhance reprogramming efficiency. Future investigations should also incorporate scVI analyses to identify molecular barriers that may impede the reprogramming process.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.